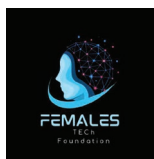


November 2024

TECH ACCESS FOR URBAN REFUGEES

Sudanese experience in Uganda



Acknowledgement

Access to Skills and Knowledge network #ASKnet is a network of nine community-driven organisations that aim to build a transformative and sustainable open society and a professional media ecosystem.

Female Technology Foundation (FTF) is a newly established organisation by female Sudanese urban refugees in Kampala dedicated to bridging the gender gap in technology, fostering innovation, and supporting women's economic and digital growth.

#ASKnet proudly supported this research study in collaboration with the r0g agency for Open Culture and Critical Transformation, backed by funding from the German Federal Ministry of Economic Cooperation and Development (BMZ). We extend our heartfelt gratitude to all the Sudanese refugees in Kampala who participated in the focus group discussion. Additionally, we acknowledge the essential contributions of Yine Yenki Nyika, Marina Modi Sokiri, and Eva Yayl Mawa Upele in curating this impactful research study. The Focus Group Discussion (FGD) was led by Sandra Aceng, who also authored the report.

Executive Summary

The Tech Access for Urban Refugees- Sudanese Experience in Uganda study provides insights into the technology access, usage, and adaptation challenges that Sudanese urban refugees face in Uganda. Refugees in Kampala, especially women, need help accessing and using technology due to a lack of familiarity with the available resources, minimal digital literacy, and cultural constraints. Financial limitations, insufficient local support, and language barriers further complicate their efforts to adapt to Uganda's digital environment. This study highlights the coping strategies and adaptive measures refugees have developed and the specific needs for technological training, affordable devices, and community support that could enhance their social and economic integration.

Key Findings

The findings show that in terms of **Technology Access and Usage**, Sudanese refugees in Kampala primarily access smartphones and, to a lesser extent, laptops, which they use for social connections, work, and learning. Common apps include WhatsApp, Instagram, and Facebook, facilitating social interaction and communication within and beyond Uganda. However, participants, especially female refugees, face difficulties with online shopping, digital payment systems, and professional networking due to a lack of knowledge about relevant platforms and tools.

Additionally, the findings further showed that gender disparities in technology access are notable, with financial dependence and motherhood affecting women's online engagement. Sudanese cultural norms also restrict women's access to technology, as some families discourage daughters from pursuing tech-related education (such as Science, Technology, Engineering, and Mathematics (STEM) subjects) or using social media. Gender-based restrictions on digital access and education further hamper women's ability to develop digital skills and opportunities.

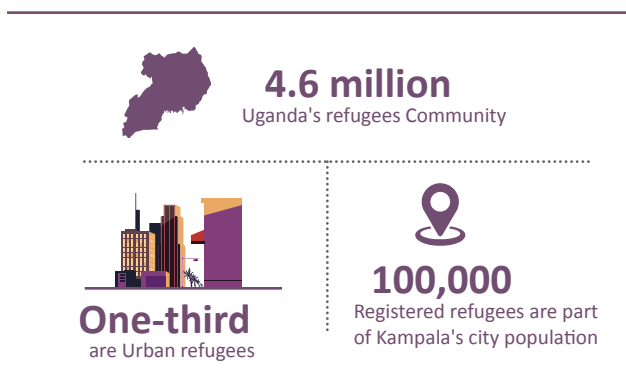
Refugees face unique challenges, such as a lack of computer training centres, language barriers, and limited knowledge of Uganda's technology and job landscape. Many have adopted self-learning and taken online courses, while some prioritise digital security training to prevent device hacking. Refugees also face economic difficulties, job instability, and a lack of motivation, which impede technological adaptation.

During the group discussion, most female participants expressed a need for foundational computer skills training, job-related digital skills, and graphic design courses to improve employment prospects. They also identified English language courses as essential for overcoming language barriers. Community support from organisations offering technology or digital literacy training, device access, and affordable internet options would significantly aid their technological adaptation.

Introduction

In Uganda, refugees significantly compose the marginalised urban population, dependent mainly on the informal sector, and are severely affected by the crisis amidst limited social protection interventions.¹ As of May 2024, Uganda hosted roughly 1.7 million refugees.²

Urban refugees are estimated to comprise about one-third of Uganda's more than 4.6 million refugees. Currently, it is believed that over 100,000 registered refugees are part of Kampala's city population.³ However, digital inequalities around mobile phone and internet access in Uganda have led to the exclusion of refugees.⁴ The increasing use of digital tools is recognized as essential for community interaction and organization, achieved through both virtual bonding and bridging.⁵ Yet, a persistent gender digital divide leads to unequal ICT access and use.



This report examines the technology access experiences of twenty-two Sudanese urban refugees in Kampala, including eighteen females and four males. It focuses on challenges refugees face regarding technology access as they adapt to their new environment and outlines potential future initiatives.

The research involved focus group discussions that gathered insights on several key topics, including technology access, the difficulties encountered by non-English-speaking urban refugees, technology usage, gender-related impacts, potential solutions, and community engagement.

The qualitative analysis highlights various scenarios and stories that illustrate the gender digital divide in technology access and its impact on Sudanese urban refugee women and men in Uganda. This exploration provides a deeper understanding of the experiences and perspectives of Sudanese urban refugees in Kampala. It emphasises their technology access challenges, usage patterns, gender-related impacts, possible solutions, and community-driven efforts to address the digital divide, particularly among refugee women.

¹ Urban Refugees' Digital Experiences and Social Connections During Covid-19 Response in Kampala, Uganda <https://www.cogitatiopress.com/mediaandcommunication/article/view/5169>

² Most common countries of origin of refugees in Uganda as of 2024 <https://www.statista.com/statistics/1241293/refugees-in-uganda-by-origin/>

³ Urban Refugees' Digital Experiences and Social Connections During Covid-19 Response in Kampala, Uganda <https://www.cogitatiopress.com/mediaandcommunication/article/view/5169/2748>

⁴ <https://www.cogitatiopress.com/mediaandcommunication/article/view/5169/2748>

⁵ Urban Refugees' Digital Experiences and Social Connections During Covid-19 Response in Kampala, Uganda <https://www.cogitatiopress.com/mediaandcommunication/article/view/5169/>

Methodology

The Female Technology Foundation, with support from the Access to Skills and Knowledge network (#ASKnet), conducted a focused group discussion on October 30, 2024, with 22 Sudanese urban refugees in Uganda, including 18 women and 4 men. This session captured insights on technology access, challenges faced by non-English-speaking urban refugees, daily technology usage, gender-related impacts, and potential solutions. Through this approach, the study gathered valuable perspectives on the specific needs and barriers within this community, exploring avenues for developing a more inclusive, accessible, and community-driven approach to technology access for urban refugees in Uganda.

Findings and Discussion

Experience that highlights a unique challenge urban Sudanese refugees face in Uganda

Access and Affordability of Devices

One significant challenge highlighted was the need for more access to devices and the digital literacy required to operate the devices. **A female refugee lawyer** explained her journey as a refugee from Sennar, Sudan, where she lost all her data after erasing her phone and arriving without devices. This loss was traumatic, as she not only lost personal memories but also faced challenges acquiring a new device due to financial constraints. The lack of technology due to her inability to purchase a new device further amplified her feeling of isolation and disempowerment in Uganda. Additionally, many female participants noted that the devices available in Uganda differ from those in Sudan, with limited familiarity with local brands and models. This gap worsens difficulties in adapting to new tools essential for integration and communication.

Digital Literacy Gaps

The need for familiarity with essential apps (e.g., Zoom, SafeBoda) often restricts job and service access. A female respondent reported having struggled with her first Zoom interview due to a lack of experience with video conferencing. However, women entrepreneurs highlighted their inability to market products on social media due to limited training effectively. Others expressed the need for tailored support in social media and e-commerce skills to create more professional business profiles and reach wider audiences.

"I just post on WhatsApp statuses that my friends see. I don't have a dedicated page for my business, and if I wanted to post on platforms like Instagram, I don't know the techniques needed."

– female refugee entrepreneur.

Gender-Based Visibility and Access Disparities

Female Sudanese refugee respondents noted that male counterparts have more visibility in certain business sectors, particularly photography. This indicates a need for a focused approach to empower women refugees with tools to equalize visibility across gender. Additionally, many female refugees lack training in developing social media content or managing digital platforms, limiting their ability to maintain a consistent online presence for their businesses.

Language and Cultural Barriers to Technology Use

Language Adaptability in Devices. The lack of Arabic support in Ugandan devices presents a significant barrier, especially for Sudanese refugees who are unfamiliar with English. An article by Cogitatio Press highlights that the lack of culturally and linguistically accessible information and services further marginalises urban refugees in their efforts to manage the pandemic. Many refugees find it difficult to read and understand information available on various media platforms, especially since most of this information is in English, a language that many are not familiar with.¹⁷

¹⁷ Hakimu Sseviiri, Amanda Alencar and Yeeko Kisira. (30 June 2022). *Urban Refugees' Digital Experiences and Social Connections During Covid-19 Response in Kampala, Uganda*. Cogitatio Press
<https://www.cogitatiopress.com/mediaandcommunication/article/view/5169/2748>

Policy Differences and Digital Access Restrictions.

Most female participants mentioned various regional access restrictions, such as the need for a VPN to access certain social media sites, such as Facebook, that have been blocked in Uganda since the 2021 presidential elections. Additionally, blocked apps and websites require these refugees to adapt to new policies that differ from those in Sudan. A report by Konrad Adenauer Stiftung indicates that Uganda's refugee policy frameworks overlook urban refugees.¹⁸

Safety, Security, and Digital Trust

The majority of female participants expressed concerns over safety in online spaces. Incidents of online scams on platforms like Jiji highlighted safety concerns, as participants shared experiences of encountering job-related fraud and feeling vulnerable to fraudulent schemes due to unfamiliarity with online security. Concerns over online safety for adolescent girls and young women, and physical safety and restrictions on mobility, are some of the main reasons behind the gender digital divide.¹⁹ Many women also faced issues determining the credibility of job postings and reported anxiety over the risk of exploitation. This gap in digital literacy and knowledge of trusted platforms leads to vulnerability in seeking employment or selling goods online.

Personal security concerns were also raised. Fear of phone and laptop theft on the streets of Kampala adds stress, with some women reluctant to carry valuable devices in public.

“If you carry your phone or laptop, it could be snatched in seconds. To grow technologically, a person needs to feel safe. Without security, it's hard to feel motivated to learn and use new technology.”

– male refugee Electrical Technician.



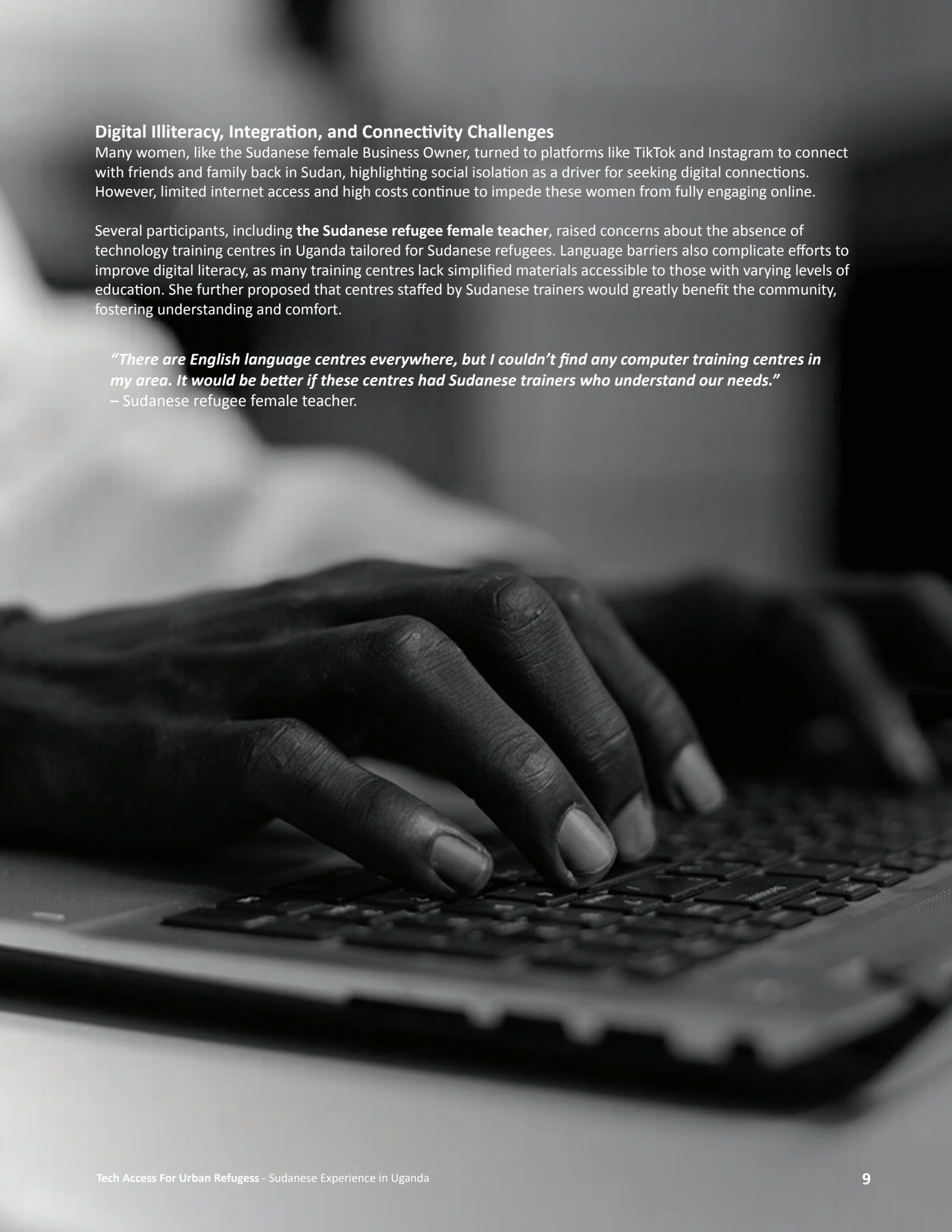
Job Market Challenges and Digital Isolation

A female interior design engineer and other participants described the frustration of being qualified but unable to secure jobs. According to the female interior design engineer, many Sudanese women refugees are well-educated but face limited job opportunities. She proposed the idea of an app dedicated to connecting refugee businesses with potential customers within the Sudanese community in Uganda, which would demonstrate the potential for community-based technological solutions to address economic challenges.

“We are all educated women, but there is no job market for us here. Many of us have different skills, but without the ability to reach people, our education and qualifications from Sudan remain unused.”
female interior design engineer.

¹⁸ Ojak Okello. (October 18, 2022). Self-Settled Urban Refugees in Kampala and Labor Market Dynamics <https://www.kas.de/en/web/uganda/veranstaltungsberichte/detail/-/content/self-settle-urban-refugees-in-kampala-and-labor-market-dynamics>

¹⁹ Susan Nicolai, Moizza Binat Sarwar, Yasser Kosbar. Digitally powered 'learning to earning' for displaced young people and adolescent girls and young women <https://www.unicef.org/media/143166/file/Digitally%20powered%20%E2%80%98learning%20to%20earning%E2%80%99.pdf>



Digital Illiteracy, Integration, and Connectivity Challenges

Many women, like the Sudanese female Business Owner, turned to platforms like TikTok and Instagram to connect with friends and family back in Sudan, highlighting social isolation as a driver for seeking digital connections. However, limited internet access and high costs continue to impede these women from fully engaging online.

Several participants, including **the Sudanese refugee female teacher**, raised concerns about the absence of technology training centres in Uganda tailored for Sudanese refugees. Language barriers also complicate efforts to improve digital literacy, as many training centres lack simplified materials accessible to those with varying levels of education. She further proposed that centres staffed by Sudanese trainers would greatly benefit the community, fostering understanding and comfort.

“There are English language centres everywhere, but I couldn’t find any computer training centres in my area. It would be better if these centres had Sudanese trainers who understand our needs.”

– Sudanese refugee female teacher.

Access to Technology

Types of technology or devices accessed by Sudanese Urban Refugees

Regarding **Access to Technology**, the findings reveal that access to technology is both a bridge for integration and a source of tension due to limitations and risks associated with device functionality, content exposure, and mental health impacts. Respondents provided diverse insights into the types of technology accessible to them and how they use these devices daily. These highlighted the enabling and constraining aspects of technology for Sudanese refugees in Uganda.

Types of Technology and Usage

Both female and male respondents described various technologies they accessed, predominantly mobile phones, laptops, and various apps that supported communication, learning, work, and leisure. Mobile devices emerged as the central technology, with platforms like **WhatsApp, Facebook, and app X (Twitter)** frequently mentioned for their role in maintaining social connections and facilitating access to current events, especially news related to Sudan. Specific apps like **Signal, Telegram, and Google Meet** were appreciated for their privacy features and lower data consumption, which aligns with the respondents' emphasis on security and cost-effective communication.

Notably, respondents utilised apps based on their functional needs, as seen in the table below.

Social media platforms and applications	How do they use the platforms daily?
 Facebook, Twitter, TikTok	Used for news, connecting with others, and stress relief.
 WhatsApp, Signal	Communication and Coordination. Essential for work-related communication, family updates, and coordinating community activities.
 LinkedIn, Coursera, Canva	Educational and Professional Apps. These were mentioned for personal development and career building, with LinkedIn playing a role in professional integration into Ugandan society.
 Google Maps, SafeBoda	Utility Apps. Used to navigate the new environment, access services, and locate educational institutions.

The daily use of these technologies indicates a high reliance on mobile and digital tools for social, educational, and work purposes, reflecting the vital role these devices play in navigating life as refugees.

Challenges with Technology Access and Usage

The discussion highlighted several barriers to technology access and usage, such as:

- **Device Limitations:** Respondents like the female Lawyer and a male electrical engineer pointed out issues with older or lower-capacity devices, which limited the range of apps they could install and use, restricting their technological engagement.
- **Digital Literacy and Knowledge Gaps:** Some respondents, such as the male electrical engineer, admitted to limited familiarity with broader technological applications, affecting their ability to fully leverage digital tools for research, learning, or professional tasks. Education and digital skills are strongly correlated, as women and girls with some secondary education are six times more likely to be online than those with only primary education.²⁰
- **Content and Data Management:** Data-intensive apps pose storage challenges, as seen when respondents' children would uninstall necessary apps to create space for games.
- **Psychological Impacts:** There was a shared sentiment regarding the mental health impact of consuming distressing news and violent imagery, especially related to the Sudan conflict. This exposure led some, like the male Electrical Engineering, to disengage from platforms like Facebook to avoid emotional strain.



²⁰ <https://www.unicef.org/media/143166/file/Digitally%20powered%20%E2%80%98learning%20to%20earning%E2%80%99.pdf>

How do Sudanese Urban Refugees use the technology or Devices Daily

Social Integration and Community Building

Technology facilitated both female and male respondents' social integration and community building in Uganda. Social media, LinkedIn, and WhatsApp groups helped them connect with Ugandan locals and professionals, participate in volunteer initiatives, and stay updated on Sudanese community affairs. As the female English language teacher noted, LinkedIn allowed her to connect with Ugandan educators and learn about the local education system, supporting her integration efforts. Social media also served as a platform for fundraising and emergency responses, demonstrating its role in community resilience and support.

Parental Concerns and Technology Use Among Children

Many respondents expressed concerns about their children's interaction with technology during the discussion. There was a shared perception that unregulated access to phones and apps could expose children to inappropriate content and limit critical thinking. Some, like the refugee female lawyer, noted differences between Sudanese and Ugandan norms regarding children's phone access, indicating a cultural consideration in balancing educational benefits with potential risks. Respondents mentioned the need for parental control features and legal awareness about restricted sites to ensure safe technology use for children.

Device Limitations and Adaptation

The female lawyer's story reflects the constraints of limited device capacity, highlighting how storage size affects app use. For instance, while she uses Signal for secure communication, her children often uninstall it to make space for other apps. This illustrates a recurring issue for refugees with limited financial means: low-capacity devices restrict their access to tools that might support professional and social integration. The female lawyer's comment, "I use apps with smaller file sizes," underlines how technology usage is often driven by device limitations rather than preference.

Social Media and Mental Health Concerns

The male electrical engineer's experience underscores the mental strain refugees face with social media. After feeling overwhelmed by "news of deaths" on Facebook, he decided to delete the app. He appreciates the connectivity and research potential technology provides in Uganda but avoids negative content. This example echoes a common theme: technology can be both a lifeline and a source of trauma as refugees navigate traumatic news, hate speech, and violent content. As the male social worker notes, "Starting your day with negative news can ruin it," emphasizing how technology use requires careful moderation for mental health. It is essential to highlight the role of digital networks in spreading disinformation, hate speech, and forms of control in the context of forced migration, which exacerbates social exclusion and vulnerability among refugees.²¹

²¹ <https://www.cogitatiopress.com/mediaandcommunication/article/view/5169/2748>



The Impact of Gender on Access to Technology

This section highlights the various social, economic, and cultural factors that shape how men and women access and experience technology. These factors collectively influence access to and usage of technology in ways that are often less favourable for women, particularly in Sudanese society and among refugee populations.

Economic and Educational Barriers

Economic constraints are a significant aspect of understanding technological access. For both men and women, financial limitations often dictate whether they can afford advanced devices or internet access, but this issue disproportionately affects women. A female data analyst notes, *“There is always a connection between technology, the economic situation, and the level of education,”* linking financial stability to the quality of technology one can afford. Additionally, financial dependency often leaves women without the means to buy devices independently, which impacts their digital literacy and career prospects.

“There is always a connection between technology, the economic situation, and the level of education,”

Educational barriers also restrict access. Many women lack technical training due to cultural biases discouraging women from entering technical fields. According to the male electrical engineer, *“The educational system in Sudan... doesn’t offer technological education until later on, like programming languages,”* a delay that leaves women less prepared for digital work environments. These educational disparities reinforce a cycle where men are more technologically proficient and better equipped to work in tech-related fields.

“The educational system in Sudan... doesn’t offer technological education until later on, like programming languages,”

Societal Expectations and Gender Roles

Cultural expectations often restrict women's ability to engage with technology freely. The findings propose a societal narrative in Sudan that limits women's roles and access to technology, particularly in settings where traditional gender roles are strongly enforced. A male social worker shared, "In Sudan, there are no women fixing phones or laptops in most places," indicating societal disapproval toward women pursuing technical work. This cultural bias extends to the household, as a female Lawyer explains, "My husband... wouldn't allow me to use the phone after 11:00 PM," which showcases how some husbands monitor and control their wives' technology use, which reinforces patriarchal norms.

Men are generally given more freedom with technology, with some encouraged to develop professional skills in this area. Women, in contrast, face cultural expectations to prioritize family and childcare, resulting in less time and fewer resources dedicated to technology use. The male Electrical Engineering also observes, "There are more challenges for refugees, especially in Uganda, such as time constraints... most of the time, they are involved in childcare," this highlights how caregiving duties impact women's ability to develop digital skills.

"There are more challenges for refugees, especially in Uganda, such as time constraints... most of the time, they are involved in childcare,"

Technological Limitations and Gendered Access

Women's roles as caregivers and homemakers often mean prioritizing family needs over personal access to technology. This situation is compounded by a lack of motivation and skills to pursue technical growth among some women, as a female human rights advocate points out: "Women often lack the desire for continuous education... this is a major issue, as economic conditions make technology less available to women." For those who do want to pursue skills, there are still barriers. For instance, gendered attitudes in society can cause fewer job opportunities in tech for women.

The lack of technological infrastructure in Sudan exacerbates these limitations. The male electrical engineer explains that "the lack of a technological environment in Sudan means that even when you have a phone, there's no electricity or internet, making it useless." This affects both men and women but compounds the challenges for women who have limited opportunities to gain experience and access resources outside the home.

: "Women often lack the desire for continuous education... this is a major issue, as economic conditions make technology less available to women."

"the lack of a technological environment in Sudan means that even when you have a phone, there's no electricity or internet, making it useless."

Resilience and Adaptive Strategies

Despite these challenges, some women have found ways to leverage their existing skills and limited access to technology to succeed professionally. A female graduate illustrates how *“many Sudanese women are advanced in technology, like Sarah Fadl, who is skilled in media.”* Women’s resilience in finding creative solutions, such as social media work, highlights their potential to excel given the opportunity. The COVID-19 pandemic provided some women with digital experience they otherwise might not have had, as the female development manager mentions: *“The lockdown led to a shift to online education... we had some experience, especially in managing children and using technology.”* Findings show that digitally enabled and mediated social networks and/or connections through bonds, bridges, and links are crucial in supporting refugees to cope with crisis effects.²²

“many Sudanese women are advanced in technology, like Sarah Fadl, who is skilled in media.”

²² <https://www.cogitatiopress.com/mediaandcommunication/article/view/5169>

Additional Challenges for Women and Men in Accessing Technology and Online Opportunities

Additional specific challenges Sudanese refugees face in accessing technology in Uganda highlight barriers related to socioeconomic status, cultural expectations, and the availability of resources. The findings reflect the layered difficulties urban refugees encounter and reveal a pronounced gap in technological access and digital literacy between men and women. Issues that emerged include financial constraints, cultural perceptions of gender roles, and the scarcity of role models to inspire tech-driven pursuits.

Financial and Structural Barriers to Technology Access

Financial limitations are a primary obstacle for refugees trying to access technology. A female lawyer reflects on how difficult it is to obtain digital devices as a refugee, explaining, *“Our opportunities to access technology are limited because most people have lost their jobs, lost their sources of livelihood, and don’t have money to save to upgrade their devices.”* In Sudan, cooperative associations enabled people to purchase devices on credit, a system that no longer exists for them in Uganda. The lack of financial resources and support networks makes it challenging to access even basic technology, further limiting refugees' economic opportunities and social mobility.

The business owner adds that refugees often “start from scratch and exert effort to get money,” underscoring the financial difficulties in adapting to life in Uganda. This lack of financial security hinders Sudanese refugees' potential to invest in technological devices and training that could otherwise support them in building livelihoods through online or tech-based work. Thus, the economic challenges of being a refugee create a persistent barrier, hindering long-term digital inclusion and professional development.

Gender-Based Constraints and Digital Security

Women face additional challenges rooted in gendered cultural norms and digital security concerns. Gender norms in both home and host countries contribute to the ongoing gender digital divide, resulting in unequal access to and use of ICT.²³ The female Sudanese with a background in Applied and Industrial Chemistry notes the self-imposed limits in Sudanese society, where women often pursue traditional fields, such as psychology or social sciences, and leave technology-focused studies to men. This cultural conditioning reinforces gender-based educational and professional divides, which can make it harder for women to develop digital skills or consider tech-related careers.

The female development manager raises an additional concern around digital security, noting that “women face more abuse than men” in online spaces, a factor that deters many women from engaging fully in digital platforms. The fear of online harassment and abuse often means that women avoid participating in digital spaces or limit their online presence, missing out on valuable learning and professional opportunities. She also further highlights that “opportunities are always fewer for women,” and the issue of wage disparity and monopolization in job networks further disadvantages women in accessing well-paying online work opportunities.

²³ <https://www.unicef.org/media/143166/file/Digitally%20powered%20%E2%80%98learning%20to%20earning%E2%80%99.pdf>

Lack of Inspiring Role Models

A recurring issue was the need for visible role models to inspire Sudanese women in technology. The male electrical engineer laments, *“There are no inspiring women... We don't find women showing other women how to access technology and share their experiences.”* This absence of role models restricts women’s vision of what is possible in tech, limiting the ambition and drive of those who might otherwise pursue digital careers.

“There are no inspiring women... We don't find women showing other women how to access technology and share their experiences.”

However, the female human rights advocate offers a counter-narrative, sharing her story of resilience and innovation. She recounts her experience in data mining and app design, noting, *“I designed an application, kind of like WhatsApp... to enhance the capabilities of these youth in civic education.”* Her achievements demonstrate that Sudanese women have the capability and skills to succeed in tech-related fields, but they often remain unseen or unacknowledged in broader discussions. The female human rights advocate’s story highlights that potential exists but is often obscured by the challenges and limitations placed upon refugees.

“I designed an application, kind of like WhatsApp... to enhance the capabilities of these youth in civic education.”

Potential for Empowerment through Technology

Despite these barriers, the data indicates a growing awareness of technology's potential for socioeconomic empowerment, particularly among younger generations. The female data analyst points out that “the new generation interacts with technology differently and understands its importance,” suggesting a generational shift in the approach to digital literacy. This younger demographic, including men and women, sees the value in digital platforms for freelancing and entrepreneurship. Encouraging this shift could be key to building a more digitally inclusive environment for refugees.

The female lawyer provides an insightful observation on the communal aspect of technological challenges, emphasising, *“The issue of women is a societal issue... we are all in a state of poverty, and if these devices were provided, I believe there would be inspiration for both men and women.”* Her comments underscore the importance of collective support and resources to create opportunities that benefit the entire community, emphasizing that when digital access is expanded, it can empower both genders to innovate and inspire.

“The issue of women is a societal issue... we are all in a state of poverty, and if these devices were provided, I believe there would be inspiration for both men and women.”

Strategies and Adaptation in Accessing Technology for Urban Refugees

The personal strategies Sudanese refugees have developed to navigate challenges related to technology access and use in Uganda based on data gathered reveals a combination of resourcefulness, self-directed learning, and adaptation to technological limitations. The common strategies used by Sudanese refugee women included leveraging available resources, finding alternative devices, continuous learning through online courses and self-teaching, and adjusting to new forms of work that rely heavily on digital skills. This adaptability reflects the resilience of refugees, who use technology not only as a means of work but also as a way to reconnect with skills they were accustomed to back in Sudan.

Coping with Limited Access to Devices and Infrastructure

Due to limited resources, refugees often lack the proper devices to complete their work effectively. The male Sudanese refugee shared his experience when his laptop stopped working mid-project: *"I tried to fix it but couldn't, and until now, it's still broken. So, I had to work with what I had. I borrowed a laptop from a friend and used it."* Unable to fix his laptop, he creatively adapted, utilizing his phone to continue his work, even though this made tasks significantly harder. This story highlights the adaptability needed when standard technological resources are inaccessible. He explains, *"I would be sitting up until 2 or 3 AM to write,"* which speaks to the added effort required when working without a suitable device.

The female development manager also emphasises the challenges of using a phone for tasks traditionally managed on a laptop, such as creating PowerPoint presentations. She states, *"It's very hard. Before, with a laptop, it was very easy. But now, there are basic things in working with PowerPoint that I can't deal with on the phone."* These insights show that many refugees face an "adapt or fall behind" situation, where alternative devices like phones become essential, even though they limit productivity.

Self-Directed Learning and Skill Development

Many Sudanese urban refugees have embraced self-learning to address skill gaps and adjust to technology needs in their new environments. The female graduate illustrates this approach, sharing, *"I spend time over two or three months. I watch YouTube, take courses, and apply what I learn."* This ongoing commitment to learning reflects a proactive approach to technology and digital literacy. The female graduate's focus on regular course-taking in her field of communication and media allowed her to help *"build a new system"* for her organization, demonstrating how continuous learning supports professional growth.

Similarly, the female graduate relies on self-teaching methods such as YouTube and consulting knowledgeable peers. She emphasizes that *"one must be continuously learning and asking knowledgeable people,"* a strategy that has helped her gain digital marketing and social media management skills. This adaptability to digital learning tools boosted her skill set and enabled her to address organizational needs in areas like digital advocacy.

A female communication and media specialist and a female refugee with an applied and industrial Chemistry focus also emphasize the value of online courses to gain new skills. A female communication and media specialist notes, *"If I face a problem, I take online courses,"* while the female refugees with an Applied and Industrial Chemistry focus highlight how the shift from traditional office work to online work during the war pushed her to develop new strategies for engaging with technology. The war, in her view, accelerated her technological learning and adaptation, underscoring how sudden life changes can prompt individuals to adopt new ways of working and problem-solving.

Adapting Skills to New Economic Realities

In Uganda, refugees have had to adapt their skills to new economic and professional environments, often requiring technology. The female business owner, who previously worked as a hotel marketing manager in Sudan, had to pivot when she arrived in Uganda. She explains, *"I learned after I arrived in Uganda because I opened my shop and had to learn how to market through websites."* Her story shows how refugees leverage their existing tech skills and adapt them to local market needs through technology.

The female interior design engineer's comment highlights that refugees often work in fields related to gender and women's issues, indicating that many feel a sense of duty to use technology to address social issues within their communities. Her statement, "There might be campaigns... about girls," suggests a desire to develop digital advocacy skills, particularly around issues that resonate with their lived experiences. This strategic approach points to how refugees use technology to connect with and empower their communities, even in new and challenging settings.

The Role of Perseverance in Technological Adaptation

Across these stories, perseverance emerges as a common trait. Refugees spend significant time learning new technologies and applying new knowledge in real-world situations, as demonstrated by Salwan's dedication to mastering complex software over months. Similarly, the female graduate's approach to continuous learning through self-guided resources like YouTube and consulting advanced peers shows how refugees blend available resources to overcome skill gaps and develop a level of digital proficiency that supports their professional goals.

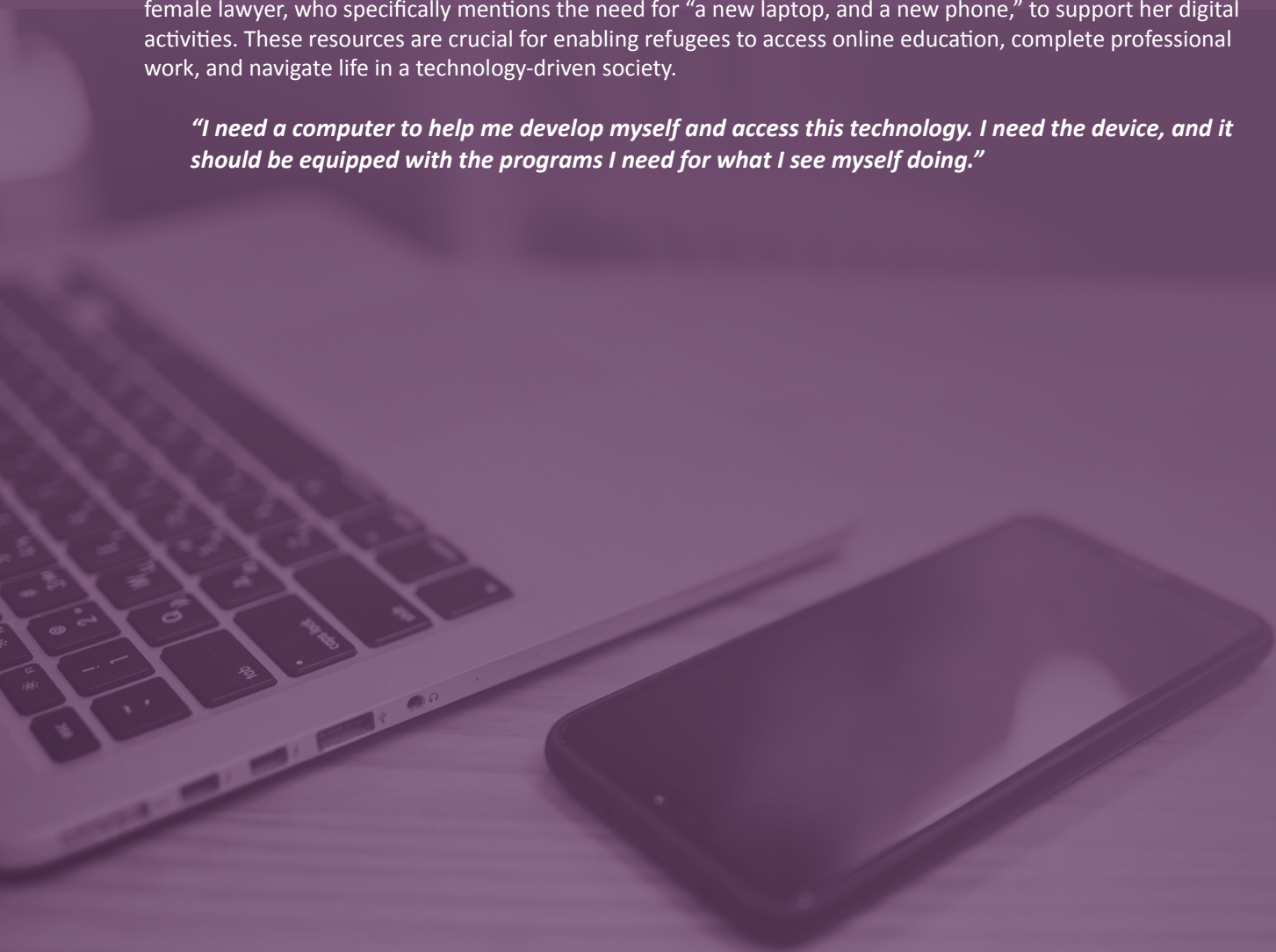
Desired Tools, Skills, and Resources for Technological Adaptation Among Urban Refugees in Uganda

In exploring the tools, skills, and resources that Sudanese urban refugees in Uganda wish they had to overcome technology-related challenges better, several of them emerged. Refugees consistently highlighted the need for material resources, comprehensive digital skills training, and an effective delivery model for technology education. Their feedback reflects a desire for structured, practical training that meets real-life needs and overcomes the gaps in short-term workshops. Refugees also expressed the importance of accessible resources, like advanced devices and software, to help them engage in meaningful work and develop skills.

Material Resources: Laptops, Phones, and Software

One of the most prominent needs expressed was access to adequate devices and software. The female development manager summarizes this priority, saying, *“I need a computer to help me develop myself and access this technology. I need the device, and it should be equipped with the programs I need for what I see myself doing.”* Refugees frequently experience barriers due to limited access to proper equipment, making even basic technological engagement challenging. The female development manager’s sentiment is echoed by others, like the female lawyer, who specifically mentions the need for “a new laptop, and a new phone,” to support her digital activities. These resources are crucial for enabling refugees to access online education, complete professional work, and navigate life in a technology-driven society.

“I need a computer to help me develop myself and access this technology. I need the device, and it should be equipped with the programs I need for what I see myself doing.”



Structured, Practical Training Over Short Workshops

Respondents expressed dissatisfaction with short-term workshops, which they found too superficial to cause lasting skills. The female human rights advocate explains, “These trainers follow up with people until they know how to write their emails... After that, it becomes like an ongoing course.” She notes that workshop models lack continuity, with people often leaving with minimal lasting knowledge. The female human rights advocate and others argue that structured, ongoing courses, as opposed to brief sessions, are essential for refugees to build sustainable digital skills.

The female human rights advocate further emphasizes the need for distinction between workshops and comprehensive courses: *“There’s a difference between doing a course and doing a workshop. You can’t bring someone who has no idea and give them a course. There’s a difference.”* Her perspective emphasizes the need for FTF to consider a longer-term, incremental approach to digital literacy, potentially through continuous education models that progressively address foundational to advanced skills.

“There’s a difference between doing a course and doing a workshop. You can’t bring someone who has no idea and give them a course. There’s a difference.”

Continuous Education for Digital Literacy and Advanced Skills

Respondents advocated for a centre for continuous education, which could provide tailored training to meet refugees’ ongoing learning needs. The female lawyer mentions, *“There should be a centre for continuous education. Technology is something I imagine could cover as many people as possible.”* This model would enable refugees to learn independently, engage in hands-on practice, and revisit complex topics. Instead of one-off workshops, continuous education programs could enable refugees to steadily develop the skills they need for daily life and work.

“There should be a centre for continuous education. Technology is something I imagine could cover as many people as possible.”

This idea of continuous education reflects a broader desire for adaptable training methods that cater to varied skill levels. It would also allow trainers to address skill gaps across different cohorts, thus minimizing repetitive training for advanced learners and providing more accessible material for beginners.

Skills Training Tailored to Real-Life Applications

Refugees emphasized that training should focus on skills directly applicable to their work and daily life. The female lawyer explains, “The research should focus on real projects related to technology, with divisions based on people’s needs.” Rather than generalized training, participants expressed a desire for digital skills specific to their professional fields, such as digital marketing for those running businesses or digital security for individuals handling sensitive information.

The male electrical engineer suggested a project called “Innovate for Refugees” to introduce refugees to skills like programming, which could offer economic opportunities in both the short and long term. He advocates for targeted skill-building that would empower refugees with technology-based skills relevant to the local context and adaptable for use globally.

Digital Security Awareness and Training

Digital security was an area of interest, especially for individuals working in advocacy or sensitive fields. A female graduate highlights the need for practical guidance, saying, “If someone wants to go back to Sudan, they should know how to secure their laptop and phone.” With security concerns a daily reality, refugees need to understand the basics of digital security and how to implement these practices in real-world situations. This includes skills like setting strong passwords, recognizing security risks, and understanding basic encryption methods. Digital security training would give them the knowledge and confidence to handle personal and sensitive information online.

Need for Self-Directed Learning Resources and Community Support

Several participants voiced support for self-learning as an effective and feasible way to acquire technology skills. A female refugee with a profession in applied and industrial chemistry notes, “*Technology is a vast sea... It’s important to know what you want to do... Everything is on the internet. You can learn anything.*” This sentiment reflects a growing interest in self-directed learning through online resources such as tutorials, forums, and free courses. However, while some refugees have succeeded with self-teaching, others suggested that periodic, instructor-led sessions should supplement self-directed learning to address gaps.

“Technology is a vast sea... It’s important to know what you want to do... Everything is on the internet. You can learn anything.”

The female interior design engineer adds that “there should be online courses, even if we come from different places, online can bring us together,” highlighting the potential for digital education to foster a sense of community. Online courses could create a shared learning environment for refugees, enabling them to troubleshoot, share insights, and support each other’s progress.

Solutions and Resources Most beneficial to Sudanese Urban Refugees and other Refugees

The finding reveals a set of prioritized needs and recommendations for technology access and integration among Sudanese urban refugees in Uganda:

Solutions and Resources

- **Skills Training and Job-Oriented Programs:** Participants highlighted the need for foundational training in computer science, data analysis, accounting, and sector-specific software commonly used in Uganda (e.g., female Data Analyst's emphasis on job-market-related programs). A female medical graduate and a female data analyst suggested these basics would equip them for immediate employment, addressing their need for applicable digital skills and job market relevance.
- **Awareness of Local Laws and Cultural Adaptation:** A female lawyer and a female development manager identified a need for legal education and cultural integration programs, including language support. Translating Ugandan technology-related laws into Arabic could foster informed community engagement and compliance, while cultural awareness training would help refugees adjust to Ugandan society.
- **Continuous, Practical Digital Education:** A female development manager and a female communication and media specialist recommended vocational training and digital skills programs that extend beyond short workshops. Their call for structured learning could be met by regular classes or community-led "train-the-trainer" models, allowing participants to build skills and expand the reach of these programs progressively.

Community and Organizational Support

- **Device Accessibility and Funding Support:** A female lawyer and agricultural machinery engineer noted that accessing technology requires financial backing and device provision. This could be achieved through partnerships with NGOs, universities, and community centres that lend or subsidize devices and provide training spaces.
- **Localized Outreach Programs:** A male social worker proposed mapping Sudanese communities in Uganda and conducting neighborhood-based programs. This could enable NGOs to address region-specific barriers and directly connect with Sudanese populations, including those in camps, who may lack basic technology.
- **Breaking Barriers in Collaboration:** A female graduate and a male social worker discussed language barriers, stereotypes, and the gap between Ugandan and Sudanese organizations. Improved communication, cultural sensitivity, and joint initiatives between Sudanese and Ugandan NGOs could build trust, reduce cultural misconceptions, and foster collaboration.

Key Challenges and Misconceptions

Respondents noted that Ugandan organizations often misunderstand the Sudanese context and cultural practices. Bridging this gap requires mutual awareness and networking, which could enhance trust and enable targeted, effective programs.

Conclusion

The analysis highlights the importance of culturally sensitive and gender-specific digital access programs in Uganda to empower Sudanese refugees, particularly women. Tailor-made training, security measures, and economic support platforms can address key obstacles such as limited device access, digital literacy gaps, and social barriers. When implemented by FTF and partner organizations, these steps will create pathways for meaningful technological inclusion, enabling refugees to overcome systemic challenges and enhance their social and economic integration in Uganda.

Recommendations

To Female Technology Foundation (FTF):

- Develop tailored digital literacy programs focusing on essential skills such as social media marketing, online security, and basic applications like Zoom to increase the employability of Sudanese refugees.
- Create a mentorship network by connecting refugees with local tech professionals for guidance and support on navigating technology.

To Local Organizations and NGOs:

- Establish community technology centres in refugee-dense neighbourhoods, focusing on affordable device access and skills training. Arabic-speaking facilitators would enhance communication and accessibility.
- Educate refugees about online safety, including securing devices and protecting personal data from hacking.
- Create platforms where refugees can connect with mentors, potential employers, and peers to share knowledge and opportunities.

To Local Authorities and Security Organizations:

- Work with refugee communities to implement security measures that protect devices and reduce theft risks, creating a safe environment for digital engagement.
- Raise awareness about Uganda's digital laws and regulations through translated resources, ensuring refugees have legal knowledge in their language.
- Set up community-based training hubs in areas with high refugee populations to provide courses on digital literacy, graphic design, and online business.
- Create funding schemes to help refugees start or expand small businesses using technology (e.g., online advertising).

To Economic Empowerment Programs:

- Develop a digital marketplace platform specifically for Sudanese refugee women, allowing them to sell products and services online. This would support economic empowerment by creating a direct line to customers.

To Government Agencies (e.g., Ministry of ICT and National Guidance, Ministry of Education and Sports)

- Partner with telecom providers to offer refugees affordable internet packages and mobile devices.
- Develop language assistance programs or apps to help refugees overcome language barriers in job search and digital communication.
- Ensure policies emphasize inclusivity, such as enabling digital training centres near refugee settlements or urban hubs.

To International Organizations (e.g., UNHCR, IOM, World Bank)

- Collaborate with local NGOs to provide comprehensive programs on basic computer skills, online safety, and labour market-related technologies.
- Fund research and initiatives to address gender-based constraints and cultural stigmas impacting women's tech access.
- Work with banks and fintech companies to simplify access to digital payment platforms for refugees, such as introducing alternatives to Visa cards.

To Private Sector (e.g., Telecom Companies, Tech Firms)

- Develop low-cost mobile plans and devices designed to meet the financial constraints of refugees.
- Partner with local organizations to provide digital insurance services to protect devices and data.
- Create programs for refugees to gain exposure and skills in tech-related fields, bridging the gap between training and employment.

To Academic Institutions:

- Offer subsidized or free online courses for refugees in data analysis, graphic design, and coding.
- Partner with organizations to design curriculums addressing specific refugee needs, including business development and online marketing.

The background of the entire page is a dark purple aerial map of a city grid. A single building in the lower-middle section of the map is highlighted in bright yellow.

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